INCH-POUND
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3 December 1990

PERFORMANCE SPECIFICATION

HEATER ASSEMBLY, COMBUSTION VEHICULAR COMPARTMENT, 60 000 BTU/HR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, testing, manufacturing, and acceptance requirements for a forced air inlet personnel heater assembly, referred to herein as "heater" (see 6.1).

1.2 Classification. Heaters will be dual or single air of the following types (see 6.4.1 and 6.4.2):

Type I	- Extended operating range.
Type II	- Standard operating range.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2540

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50271	- Plate, Identification, Instruction and Marking, Blank.
A-A-52557	- Fuel Oil, Diesel; For Posts, Camps And Stations.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION

Federal Motor Vehicle Safety Standards (FMVSS)
Federal Motor Carrier Safety Requirements (FMCSR)

(Copies of the Department of Transportation documents are available from the U.S. Government Printing Office, Superintendent of Documents, Washington, DC 20402-9328.)

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NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

11669489	- Heater, Vehicular Compartment (Interface).
11669490	- Heater, Vehicular Compartment (Interface).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4	- Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.
ASTM D2156	- Smoke Density in Flue Gases from Burning Distillate Fuels, Test for.

(Application for copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN WELDING SOCIETY (AWS)

AWS B2.1	- Standard for Welding Procedure and Performance Qualification (Replaces AWS D10.9).

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(Application for copies may be obtained from the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials shall be in accordance with the manufacturer's material specifications for heaters. Materials shall be corrosion resistant or treated to prevent corrosion. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.5.1). Recycled materials is acceptable when processed to make new.

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, the use of dissimilar metals in metal to metal contact shall be avoided (see 4.5.1).

3.2.2 Steel components. Non-painted steel components, other than stainless steel, shall be zinc plated according to ASTM B633, type II class Fe/Zn 13 or equivalent. Heat radiating surfaces may be aluminized steel (see 4.5.1).

3.2.3 Aluminum components. Aluminum components shall be protected by chemical conversion coating which shall provide the components with the maximum corrosion resitance as well as optimum paint adhesion (see 4.5.1).

3.2.4 Identification plates. Nameplates shall be furnished in one of the configurations as specified below (see 4.5.1):

a. Material and construction shall conform to A-A-50271. Nameplate shall be permanently attached with metal fasteners.
b. Adhesive backed aluminum foil, except backing sheet, which may be made of paper, shall be removable intact by stripping without use of water or other solvents. Plates shall have an average adhesion, over bare and painted metal surfaces, of not less than 50 ounces per inch (oz/in.) of width at 77 $\pm$5 degrees Fahrenheit (°F).

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3.2.5 Welding. All heater parts that are exposed to combustion gases during normal operation shall be welded, as applicable, in accordance with AWS B2.1 (see 4.5.1).

3.2.6 Sealant. All sealing threaded fittings in the heater's fuel circuit or threads utilizing a locking compound shall be sealed or locked, with a suitable single component compound. Compound's grade shall be selected so that its strength shall not result in part damage upon disassembly for repair (see 4.5.1).

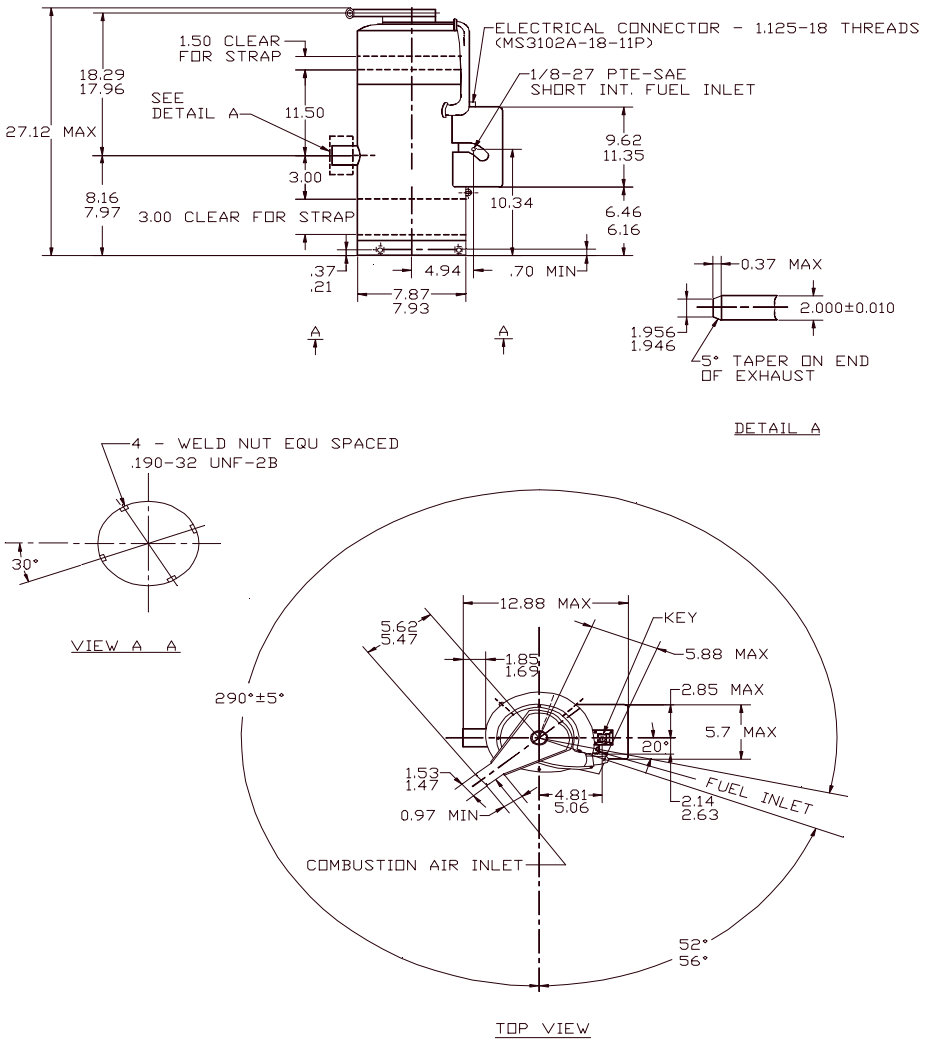
3.2.7 Prohibited materials. Asbestos, cadmium, and radioactive materials shall not be used in this item. Radioactive material is defined by Code of Federal Regulations (CFR), Title 10, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries (see 4.5.1).

3.3 Design and construction. The design and construction of the heaters shall be in accordance with heater manufacturer's standards. Heaters shall conform to the envelope dimensions and interfaces as shown on the applicable drawing (see table I) or in figures 1 or 2 as applicable (see 4.4.2, 4.5.1, 4.5.2 and 6.2).

TABLE I. Applicable drawings.

Type	Drawing number	Air inlets
I	11669489	Dual
II	11669490	Dual

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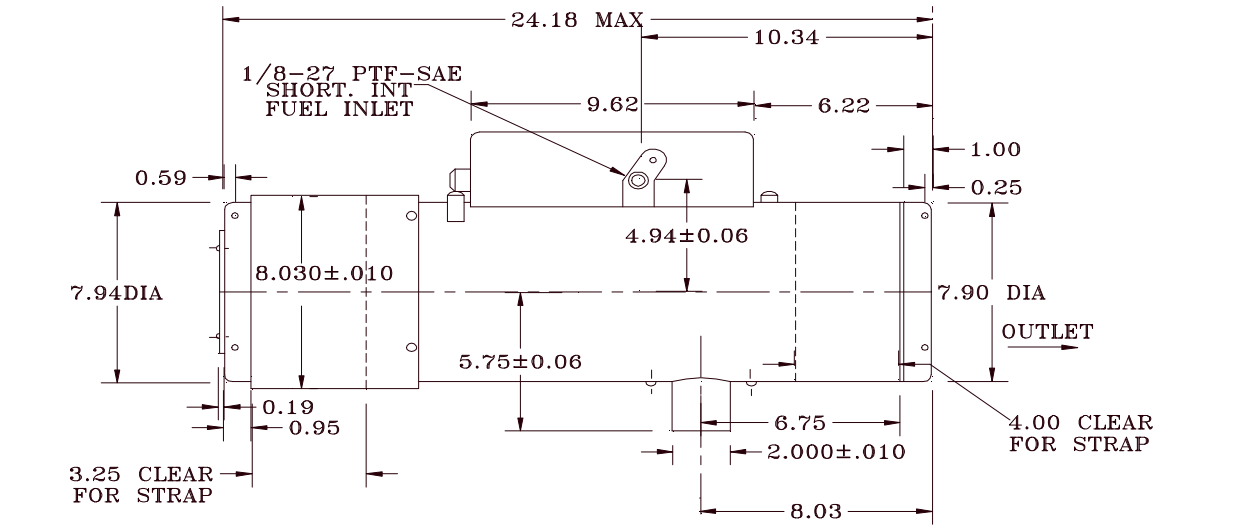
Part Identifying Number (PIN)	Prior MS part number
M62315-D	MS52288

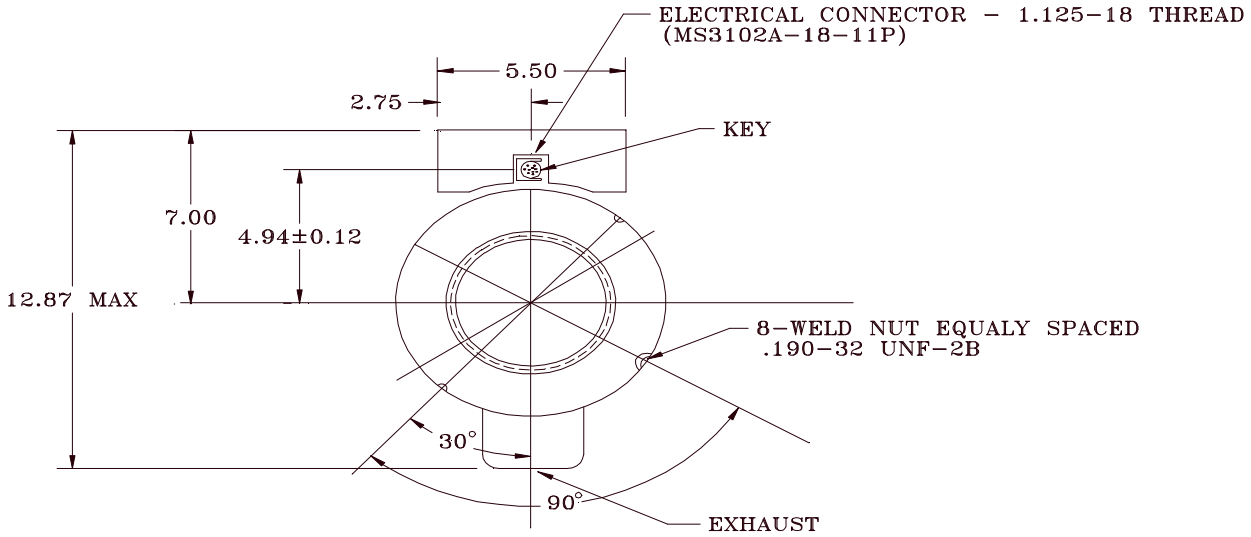
NOTES:

1. Combustion air inlet must be rotatable to any angle within the 290° range and securable in any angle such that shock or vibration shall not result in a change of position.
2. A minimum installation clearance of 4 inches at the air inlet to the blower must be provided. A minimum clearance of 3 inches above the heater cover is required for servicing.
3. Dimensions are in inches. Unless noted, tolerances are: angular ±1°, 2 place decimal ±0.06 inches.

FIGURE 1. Heater assembly, combustion dual air type.

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Part Identifying Number (PIN)	Prior MS part number
M62315-S	MS52287

NOTES:

1. A minimum installation clearance of 3 inches at the air inlet to the blower must be provided. A minimum clearance of 3 inches above the heater cover is required for servicing.
2. Dimensions are in inches. Unless noted, tolerances are: angular ±1°, 2 place decimal ±0.06 inches.

FIGURE 2. Heater assembly, combustion single air type.

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3.3.1 Weight. Weight of the heater shall not exceed 38 pounds (lb) (see 4.5.3).

3.3.2 Electrical components and wiring. The heater's electrical components and wiring shall operate and meet all environmental requirements. The wire and insulation shall be flexible to -70°F and be sized in accordance with the required rating (see 4.5.1).

3.3.3 Interchangeability. The assembly shall be so designed and fabricated that all component assemblies and parts are functionally and dimensionally interchangeable with like component assemblies and parts previously furnished by the same manufacturer. Construction shall be such that any part, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced, and adjusted without requiring modification (see 4.5.1).

3.3.4 Safety. Hot spots that could create contact burn to personnel shall be well insulated or guarded, and must fall within the area identified as "hot spot" by the applicable drawing(s). The heater shall be so designed that, with proper installation, the heater exhaust gases cannot enter the vehicle personnel compartment. The heater shall comply with all FSCSR and FMVSS requirements applicable to this item at the time of manufacture. All rotating or moving parts, hot surfaces, electrically energized or high pressure components, or other inherently hazardous items shall be effectively guarded or insulated to protect operating and maintenance personnel (see 4.5.1).

3.3.4.1 Noise level. Noise levels shall be less than 85 dB(A) at the operator's positions and other occasionally occupied positions as measured during typical operating conditions and environments.

3.4 Performance. The heater shall meet the performance requirements specified herein when mounted in any position from horizontal to vertical with the air inlet level with, or higher than, the air outlet; and, except as otherwise specified, when operating in both the high and low heat modes. The heater shall perform as specified using fuel conforming to A-A-52557, grades 1-D and 2-D and arctic grade (JP-8) aviation fuel (see 6.5).

3.4.1 Fuel.

3.4.1.1 Fuel consumption. With any fuel pressure between 3 and 15 pounds per square inch (psig) at the fuel valve inlet, fuel consumption shall not exceed the following (see 4.5.4.1):

High heat	0.092 pounds per minute (lb/min)
Low heat	0.052 lb/min

3.4.1.2 Fuel leakage. There shall be no visible leakage from the heater fuel components when 15 psig fuel pressure is applied to the inlet and the heater is not operating (see 4.5.4.2).

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3.4.2 Voltage. The heater shall start and except where otherwise specified, meet performance requirements with an applied voltage of 19 to 30 volts (V) direct current (dc) (see 4.5.5).

3.4.3 Current. At 24 Vdc, the current draw from the power source shall not exceed 16 amperes (A) starting and 15 A running when the fuel preheaters are not activated. Current draws may increase by 5 A when the fuel preheaters or start circuit are activated at approximately 45°F (see 4.5.6).

3.4.4 Heat output. With a ventilating back pressure of 1.30 inch (in.) of water and an exhaust pressure of 0.95 (type I) or 0.85 (type II) in. of water set as specified (see 3.4.5), the heater output shall be as follows (see 4.5.7):

High heat	19 to 30 Vdc, 60 000 British thermal units per hour (Btu/hr) ±10 percent.
Low heat	19 to 30 Vdc, 30 000 Btu/hr ±20 percent.

3.4.5 Air discharge rate. With 24 Vdc applied, the ventilating air discharge rate, based on a standard air density of 0.075 pounds per cubic foot (lb/ft^3) at 29.92 in. of mercury (Hg), 70°F, into the specified restriction shall be as follows (see 4.5.8):

Back pressure Minimum (inches of water) ventilation	Exhaust (inches of water)	Discharge rate Minimum (standard ft^3/min scfm)
	Type I	
1.30	0.95	165
	Type II	
0.00	0.00	200
1.30	0.85	150

3.4.6 Overheat switch. When subjected to the specified test, the heater shall meet the requirements for the applicable type (see 4.5.9):

a. Type I. The heater shall be equipped with an overheat switch to reduce the fuel flow to control the temperature of the discharge air so that it falls between 275 and 350°F. When the air restriction is removed the heater shall continue burning without requiring a return to the start cycle. Upon further restriction of air flow to not greater than 75 scfm the ventilating air temperature shall not raise above 475°F and the heater shall shut off its fuel flow then enter its purge cycle and shut itself off.

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b. Type II. The heater shall be equipped with an overheat switch and an automatic fuel shut off device which will automatically open a circuit and shut off the fuel flow when the temperature of the discharge air reaches 400°F. Subsequently, it shall enter its purge cycle and then shut itself off.

3.4.7 Smoke density. After the heater has been operating on high heat for 5 minutes, the exhaust gases shall not exceed a smoke spot number of three below 55°F and five above 55°F as defined in ASTM D2156 (see 4.5.10).

3.4.8 Starting time. The starting time, measured from the time the heater is turned on until the igniter circuit is de-energized shall not exceed the following for the specified types (see 4.5.11):

Type I	- Two minutes for temperatures above -20°F. Four minutes for temperatures below -20°F.
Type II	- Four minutes at all temperatures.

3.4.9 Purging time. The purging time for both type I and II, measured from the time the heater switch is turned off until the blowers shut off at which time the exhaust temperature shall not be greater than 350°F, shall not exceed 4 minutes (see 4.5.12).

3.4.10 Toxic fumes. The heater shall be designed so that the combustion products do not contaminate the ventilating air during operation of the heater (see 4.5.13).

3.4.11 Endurance. The heater shall meet the requirements of 3.4.1, 3.4.2 through 3.4.7 and 3.4.10 after 800 hours of operation, and 3.4.1.2, 3.4.8 and 3.4.9 during 800 hours of operation; including at least 2400 start cycles. The following maintenance is permissible (see 4.5.14):

a. Ice may be removed from the heater fuel system.
b. Cleaning and adjustment of the flame detector switch, but not more than two times.
c. Replacement of the igniter, but not more than one time.
d. Replacement of the burner assembly if required, but no more than frequently than once per sample during the test procedure.

3.4.12 Environmental conditions. The heater shall meet the requirements of 3.4.1 through 3.4.10 after being subjected to the environmental conditions specified in 3.4.12.1 through 3.4.12.6.

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3.4.12.1 High temperature. The heater shall meet the requirements after storage at 160°F and during operation at temperatures up to 75°F (see 4.5.15.1).

3.4.12.2 Low temperature. The heater shall meet the requirements after storage at a temperature of -65°F and during operation at -55°F provided the fuel is in liquid form (see 4.5.15.2).

3.4.12.3 Humidity. The heater shall meet the requirements after exposure to relative humidity up to 100 percent (see 4.5.15.3).

3.4.12.4 Salt fog. The heater shall meet the requirements after exposure to salt fog for 96 hours (see 4.5.15.4).

3.4.12.5 Vibration. The heater shall meet the requirements during and after sinusoidal vibration in accordance with figure 3 herein. The input is to be measured at the interface of the heater and support bracket (see 4.5.15.5).

3.4.12.6 Shock. The heater shall meet the requirements after being subjected to 40 gravity units (g) shock loading in three principal axes (see figure 4). The input is to be measured at the interface of the heater and support bracket (see 4.5.15.6).

3.4.12.7 Electromagnetic interference (EMI). The heater shall meet the following EMI narrowband requirements for all operating modes including the start cycle. However, the single switching actions which initiates and ends, respectively, the heater start cycle, do not have to meet the EMI requirements (see 4.5.15.7).

a. Electromagnetic emissions in the frequency of range of 10 kilohertz (kHz) to 50 millihertz (MHz) shall not appear on power leads in excess of the values shown in figure 5.
b. Field emissions in the frequency range of 14 kHz to 10 gigahertz (GHz) shall not be generated and radiated in excess of the values shown in figure 6 at a test distance of 2 meters (m). In the frequency range of 30 MHz to 10 GHz, the limits shall be met for both horizontally and vertically polarized waves.

3.5 Protective coating. All steel parts, except stainless steel, that are subject to temperatures exceeding 300°F, shall be coated forest green or olive drab, using a suitable heat resisting paint. All other painted steel parts shall be coated as described above or coated suitably to conform to the requirements specified herein. All painted surfaces shall be suitably cleaned and primed prior to painting. Unless otherwise specified, final topcoat color shall be camouflage

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green (see 6.6). When specified by the procuring activity (see 6.2), the final top coat color shall be white interior military vehicle paint. If Chemical Agent Resistant Coating (CARC) is required, it shall be specified in the contract or order (see 6.2).

3.6 Identification marking. Marking shall be permanent and legible. The identification marking, as a minimum, shall include (see 4.5.2 and 6.2):

a. Department or agency identification
b. Registration number
c. Legend as specified for the appropriate military agency
d. Manufacturers identification (CAGE)
e. National Stock Number (NSN)
f. Contract number
g. Part or Identifying Number (PIN)

3.7 Workmanship. The heater shall be free of defects such as burrs, scratches, chips, sharp edges, corrosion, scale and dirt (see 4.4.2 and 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspections (see 4.4).
    1. Examination (see 4.4.2).
    2. Acceptance tests (see 4.4.3).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 $\pm$18°F.
b. Barometric pressure: 28.5 (+2, - 3) inches Hg of mercury.
c. Relative humidity: 50 $\pm$40 percent.

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on the first and two additional samples as specified when a first article sample is required (see 3.1).

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TABLE II. Classification of inspections.

Title	Require-ment	Inspec-tion	First article samples			Conformance	
			1	2	3	Exam-ination	100% Acceptance test
Heater position 1/			H	V	H		
Materials and construction	3.2 thru 3.2.4 and 3.3.2 thru 3.3.4	4.5.1	X	X	X		
Defects (see table III)	3.3, 3.5, 3.6 and 3.7	4.5.2	X	X	X	X	
Weight	3.3.1	4.5.3	X	X	X		
Fuel consumption	3.4.1.1	4.5.4.1	X	X	X		X
Fuel leakage	3.4.1.2	4.5.4.2	X	X	X		X
Voltage	3.4.2	4.5.5	X	X	X		X
Current	3.4.3	4.5.6	X	X	X		X
Heat output	3.4.4	4.5.7	X	X	X		
Air discharge rate	3.4.5	4.5.8	X	X	X		X
Overheat switch	3.4.6	4.5.9	X	X	X		X
Smoke density	3.4.7	4.5.10	X	X	X		X
Starting time	3.4.8	4.5.11	X	X	X		X
Purging time	3.4.9	4.5.12	X	X	X		X
Toxic fumes	3.4.10	4.5.13	X	X	X		
Endurance	3.4.11	4.5.14	X	X			
High temperature	3.4.12.1	4.5.15.1	X	X	X		
Low temperature	3.4.12.2	4.5.15.2	X	X	X		
Humidity	3.4.12.3	4.5.15.3		X			
Salt fog	3.4.12.4	4.5.15.4	X				
Vibration	3.4.12.5	4.5.15.5		X	X		
Shock	3.4.12.6	4.5.15.6	X	X			
EMI	3.4.12.7	4.5.15.7		X	X		

1/ Heater position: H = Horizontal, V= Vertical.

4.4 Conformance inspection. Conformance inspection shall include the examination of 4.4.2 and the test of 4.5.4.1 through 4.5.1.2 (see table II).

4.4.1 Sampling. Samples for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4.

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4.4.2 Examination. Each heater shall be examined for compliance with the requirements specified in 3.3 and 3.5 through 3.7. Defects shall be classified as listed in table III. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Identification marking, improper (see 3.6).	Visual
104	Faulty workmanship, affecting performance (see 3.7).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
202	Protective coating, improper (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.3 Acceptance tests (100 percent). Each heater shall be subjected to the acceptance test specified in table II.

4.5 Methods of inspection. Except as otherwise specified herein, performance tests shall be conducted with the heater mounted in the positions specified in table II, at both high and low heat, and at input voltages of 19 ±0.5 Vdc, 24 ±0.5 Vdc and 30 ±0.5 Vdc. Acceptance testing shall be conducted at 24 ±0.5 Vdc only and with the heater in the vertical position, except for the fuel consumption test which shall be conducted in both positions. The fuel shall be as specified in 3.4 and the fuel pressure shall be 8 ±1 psig. Prior to each start, the heater shall be cooled for at least 10 minutes.

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4.5.1 Materials and construction. To determine conformance to 3.2 through 3.3, and 3.3.2 through 3.3.4, inspection and material welding certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3, 3.5, 3.6, and 3.7, shall be determined by examination for the defects listed in table III. Examination shall be visible, tactile, or by measurement with standard inspection equipment.

4.5.3 Weight. To determine conformance to 3.3.1, the heaters shall be weighed.

4.5.4 Fuel.

4.5.4.1 Fuel consumption. To determine conformance to 3.4.1.1, the fuel flow shall be measured with a fuel pressure of 15 (+0, -1) psig applied to the fuel valve inlet and subsequently with a pressure of 3 (+1, -0) psig, similarly applied.

4.5.4.2 Fuel leakage. To determine conformance to 3.4.1.2, the heater shall be operated for at least 5 minutes at a fuel pressure of 14 $\pm$1 psig. Subsequently, there shall be no evidence of fuel leakage.

4.5.5 Voltage. To determine conformance to 3.4.2, applied voltage shall be monitored throughout performance testing.

4.5.6 Current. To determine conformance to 3.4.3, current draw from the power source shall be monitored during performance testing. Running current shall be measured after the igniter or start circuit is de-energized.

4.5.7 Heat output. With ventilating and exhaust back pressures set as specified (see 4.5.8), thermocouples measuring input and output air temperatures, and measuring ventilating air flow (see 4.5.8), the heater shall be operated until the ventilating outlet temperature has stabilized. Output thermocouples shall be located 42 $\pm$1 in. from the heater, within the duct used in 4.5.8. When testing is performed at ambient temperatures above 55°F, and the overheat switch cycles, the switch may be bypassed. Bypassing is only allowed when the outlet temperature reaches between 275 and 350°F (type I heaters) or 400 $\pm$75°F (type II heaters). Heat output shall be calculated, using the following equation, to determine conformance to 3.4.4.

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H = V x d x Cp x T
H = Heat output in Btu/hr
V = Air flow in cubic feet per hour (ft^3/hr) (standard air)
d = Density of air at 70°F, 0.0750 lb/ft^3 (standard air)
Cp = Specific heat of air, 0.24 Btu per lb (Btu/lb) per °F, as corrected for moisture content and temperature
T = Outlet temperature, in °F, minus inlet temperature, in °F.

4.5.8 Air discharge rate. To determine conformance to 3.4.5, operate the heater as necessary and set the ventilating air restriction to produce a minimum of 1.30 inches of water at 174 ±5 scfm air flow. Measure the air flow and pressure drops. With the heater operating at 24 Vdc input, set the exhaust restriction as specified and measure the ventilating air flow rate and back pressures of the exhaust and ventilating air as specified. During subsequent testing no changes shall be made in these restrictions. The ventilating air duct shall be insulated with at least 1.5 in. of fiberglass.

4.5.9 Overheat switch. To determine conformance to 3.4.6, expose the heater to a temperature between 55 and 91°F and operate it at high heat into the duct used in 4.5.8, restricting the ventilating air flow as necessary to cause a temperature buildup in the duct. Measure the temperature of the ventilating air at a point 42 ±1 in. from the heater, within the duct. The overheat switch shall break the circuit and reduce or cut off the fuel when the air in the duct reaches the overheat temperature specified for its type. The types shall function as follows:

a. Type I. Over-heat temperatures is between 275 and 350°F. After actuation of the over heat function, the heater shall be operated for at least two minutes and measure the control temperature. Upon removing the restriction the heater shall resume burning without requiring return to the start cycle. Subsequently, the ventilating air shall be further restricted and the heater's performance shall be monitored for conformance.
b. Type II. Over-heat temperature is 400°F. The heater shall be monitored for proper execution of overheat functions and will be considered to meet the requirement if the functions are properly performed when discharge air is 400 ±75°F.

NOTE: Heater radiation effects should be negated by appropriate shielding of heat sensing devices.

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4.5.10 Smoke density. To determine conformance to 3.4.7, using the test methods set forth in ASTM D2156, measure the smoke density of the heater exhaust gases after 5 minutes on high heat. The smoke density shall not exceed a smoke number of three at a temperature below 55°F and five at a temperature above 55°F at 24 Vdc.

4.5.11 Starting time. To determine conformance to 3.4.8, measure and record the time from the starting switch being turned on until the igniter, start circuit, or preheaters are de-energized. The elapsed time shall not exceed the value specified herein. This test shall be conducted at low heat only.

4.5.12 Purging time. To determine conformance to 3.4.9, the heater's ventilating air and exhaust temperatures shall be stabilized at high heat and the starting switch shall be turned off. Measure the exhaust temperature for both type I and II (not more than 25 in. from the heater's exhaust tube's end) and the time from the starting switch being turned off until the blowers shut off. The elapsed time shall not exceed 4 minutes and exhaust temperature not more than 350°F.

4.5.13 Toxic fumes.

4.5.13.1 Single air intake heaters. Prior to heater assembly, all heat exchangers shall be tested for leakage with 10 psig applied under water. No leakage shall be permitted. The single air intake heater need only be tested again at the completion of the first article, at which time leakage shall not exceed 0.5 scfm with 3 in. water pressure applied.

4.5.13.2 Dual air intake heaters. Prior to heater assembly, all heat exchangers shall be tested for leakage with 10 psig applied under water. No leakage shall be permitted. Once heater is assembled, leakage shall not exceed one scfm with the combustion air inlet sealed and 3 in. water pressure applied to the exhaust outlet.

4.5.14 Endurance. To determine conformance to 3.4.11, the heater shall be operated with the specified restrictions (see 4.5.8) for a total of 800 hours "ON" time (ignite and run). Operation shall be in 30 minute cycles consisting of 20 minutes "ON" time and 10 minutes "OFF" time (purge and cool). The total 800 hours of operation shall consist of only "ON" time. "OFF" time will not be included in compilation of operating time. During endurance testing the heater shall be tested as specified by 4.5.11 and 4.5.12. After endurance testing the heater shall be tested as specified by 4.5.13. Operation shall consist of the following phases:

PHASE I. Using 2-D fuel at heater ambient temperature of 73 $\pm$18°F.

a. Set controls at low heat and operate heater in cycles previously described for 8 hours with an input voltage of 19 $\pm$0.5 Vdc.
b. Repeat step a using an input voltage of 24 $\pm$0.5 Vdc.
c. Repeat step a using an input voltage of 28 $\pm$0.5 Vdc.

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d. Reset temperature control to high heat and repeat steps a, b and c.
e. Repeat steps “a” through “d” until 600 thirty minute cycles are completed (200 “ON” hours).

PHASE II. Using 1-D fuel at -25 ±5°F (both fuel and heater) repeat steps “a” through “e”, except 300 thirty minute cycles (100 “ON” hours) shall be completed and in Step “c” substitute 30 ±0.5 Vdc in lieu of 28 ±0.5 Vdc.

PHASE III. Using JP-8 fuel at -55 ±5°F (both fuel and heater) repeat steps “a” through “e”, except 300 thirty minute cycles (100 “ON” hours) shall be completed and in Step “c” substitute 30 ±0.5 Vdc in lieu of 28 ±5°F Vdc.

PHASE IV. Repeat phases I through III for an additional 400 “ON” hours. After the final cycle at each specified temperature and still at that temperature the heater shall be tested as specified in 4.5.4 through 4.5.10 and 4.5.12.

4.5.15 Environmental. After the completion of environmental testing as specified in 4.5.15.1 through 4.5.15.6, the heater shall be subjected to the tests specified in 4.5.4 through 4.5.13.

4.5.15.1 High temperature. To determine conformance to 3.4.12.1, the heater shall be tested in accordance with MIL-STD-810, method 501, procedure I. The heater, positioned as specified in table II, shall be stored at 160°F for 24 hours.

4.5.15.2 Low temperature. To determine conformance to 3.4.12.2, the heater shall be tested in accordance with MIL-STD-810, method 502, procedures I and II as follows:

a. Procedure I. The heater, using JP-8 fuel, shall be positioned as specified in table II. The chamber air temperature shall be maintained at -65°F for at least 12 hours. The heater shall immediately be tested as specified in procedure II.
b. Procedure II. With the temperature at -65°F, the heater shall be tested as specified in 4.5.4 through 4.5.12, omitting 4.5.4.2 and 4.5.9. Prior to testing, the heater shall be stabilized at the low operation temperature. During testing, stabilize combustion for 30 minutes on low heat and again on high heat. Subsequently, testing as specified in procedure II, shall be repeated at the low operation temperature of -25°F using 1-D fuel and at +20°F using 2-D fuel. At each of the three temperatures, the tests specified in 4.5.11 and 4.5.12 shall be repeated twice more, for a total of three times. A baffle may be used to reduce cold air blasts effects for all cold room tests. The baffle must allow cold air flow around the heater. The combustion air inlet in multiport air heaters must be outside the baffle.

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4.5.15.3 Humidity. To determine conformance to 3.4.12.3, the heater shall be operated for a minimum of one hour at high heat and then tested as specified in MIL-STD-810, method 507, procedure III.

4.5.15.4 Salt fog. To determine conformance to 3.4.12.4, the heater shall be operated for a minimum of one hour at high heat and then tested in accordance with MIL-STD-810, method 509, procedure I. The heater shall be exposed to salt fog for a period of 96 hours.

4.5.15.5 Vibration. To determine conformance to 3.4.12.5, the heater exposed to temperatures between 55 and 91°F shall be rigidly mounted and subjected to vibration as specified in MIL-STD-810, method 514, procedure VIII, using the curve shown by figure 3 herein. During testing, heater shall be started, purged, operated at both heats, and monitored for intermittent operation. A resonant frequency search shall be conducted along each axis, with the heater operating at low heat, over a range of 5 to 500 Hertz (Hz). The test schedule shall include 13.333 minutes dwell time at each resonance to a maximum of four. After determining resonance at low heat, the specified resonance dwell and sinusoidal cycling tests shall be conducted. The total cycling time per axis shall be 80 minutes including the resonance dwell time.

4.5.15.6 Shock. To determine conformance to 3.4.12.6, the heater, with no fuel applied, shall be subjected to the half-sine shock pulse wave form shown by figure 4 herein according to the method specified in MIL-STD-810, method 516, procedure I. The pulse shall have an amplitude of 40 g's and a duration of 18 milliseconds (ms). During testing, the heater shall be subjected to input power and monitored for intermittent operation. Steps 2 and 5 may be omitted.

4.5.15.7 Electromagnetic interference. The heater shall be subject to the tests specified by MIL-STD-462.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

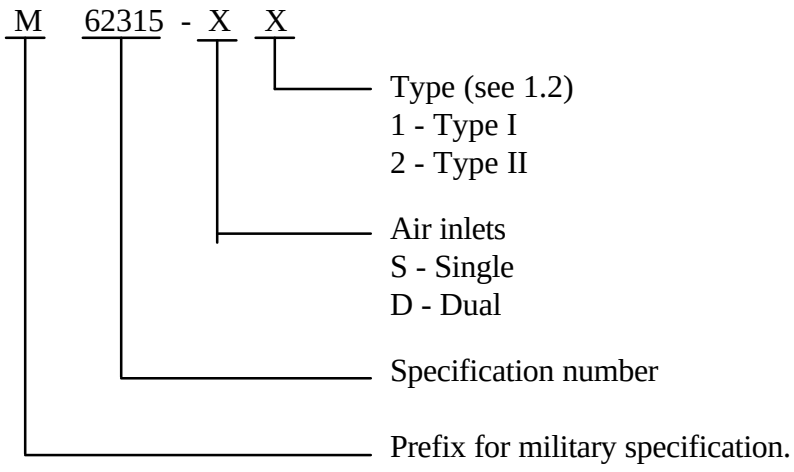
(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The heater furnished under this specification is intended to supply warm air to the crew compartment of military vehicles. Type I heaters may be substituted for type II heaters. Type II heaters will not be substituted for type I heaters. Heaters are used on M992 Series FAASV and M109 Series SPH.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of heater required (see 1.2).
c. Whether single air or dual air configuration or if either configuration is acceptable (see 1.2).
d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
e. If first article samples are specified (see 3.1).
f. Title, number and date of applicable drawing (see 3.3).
g. If final top coat color is other than as specified (see 3.5).
h. If CARC paint is required (see 3.5).
i. Pin and quantity items required (see 3.6 and 6.3).
j. Identification of appropriate service for marking (see 3.6).
k. If inspection conditions are other than as specified (see 4.2).
l. If first article inspection is not required (see 4.3).
m. Packaging requirements (see 5.1).

6.3 Part or identifying number (PIN). The PIN to be used for heater assembly acquired to this document are created as follows:



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6.4 Definitions.

6.4.1 Dual air heaters. Dual air heaters are heaters with separate air inlets for ventilating air and combustion air.

6.4.2 Single air heaters. Single air heaters are heaters with a common air inlet for ventilating air and combustion air.

6.5 Arctic grade aviation turbine fuel. Arctic grade aviation turbine fuel (JP-8) is similar to commercial aviation fuel, but has special additives for arctic conditions as detailed in MIL-T-83133 (see 3.4).

6.6 Final topcoat color. Camouflage green as used by the military conforms to color chip No. 34094 of FED-std-595 (see 3.5).

6.7 Subject term (key word) listing.

Air discharge rate
Environmental conditions
Fuel consumption
Heat output
Protective coating
Overheat switch
Smoke density
Toxic fumes

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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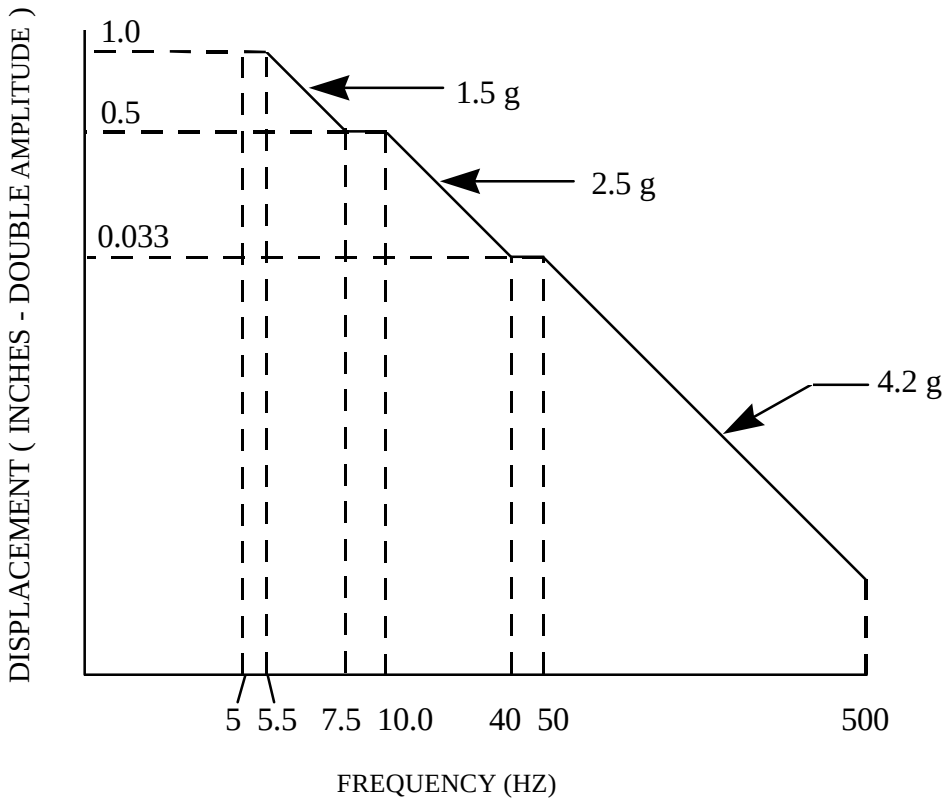
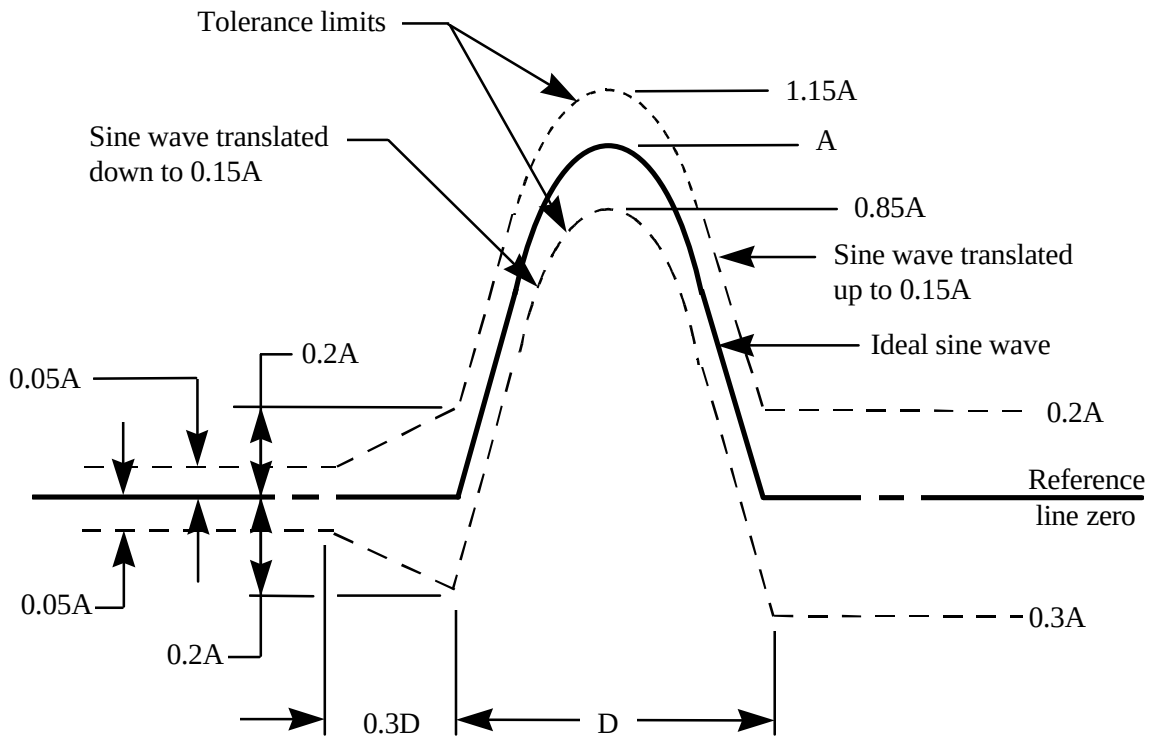


FIGURE 3. Vibration curve.

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NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The acceleration amplitude of the ideal half-sine pulse is a and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits $V_i \pm 0.1\ V_i$ where V_i is the velocity-change associated with the ideal pulse which equals $2\ AD/\pi$. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 4. Half-sine shock pulse configuration and its limits.

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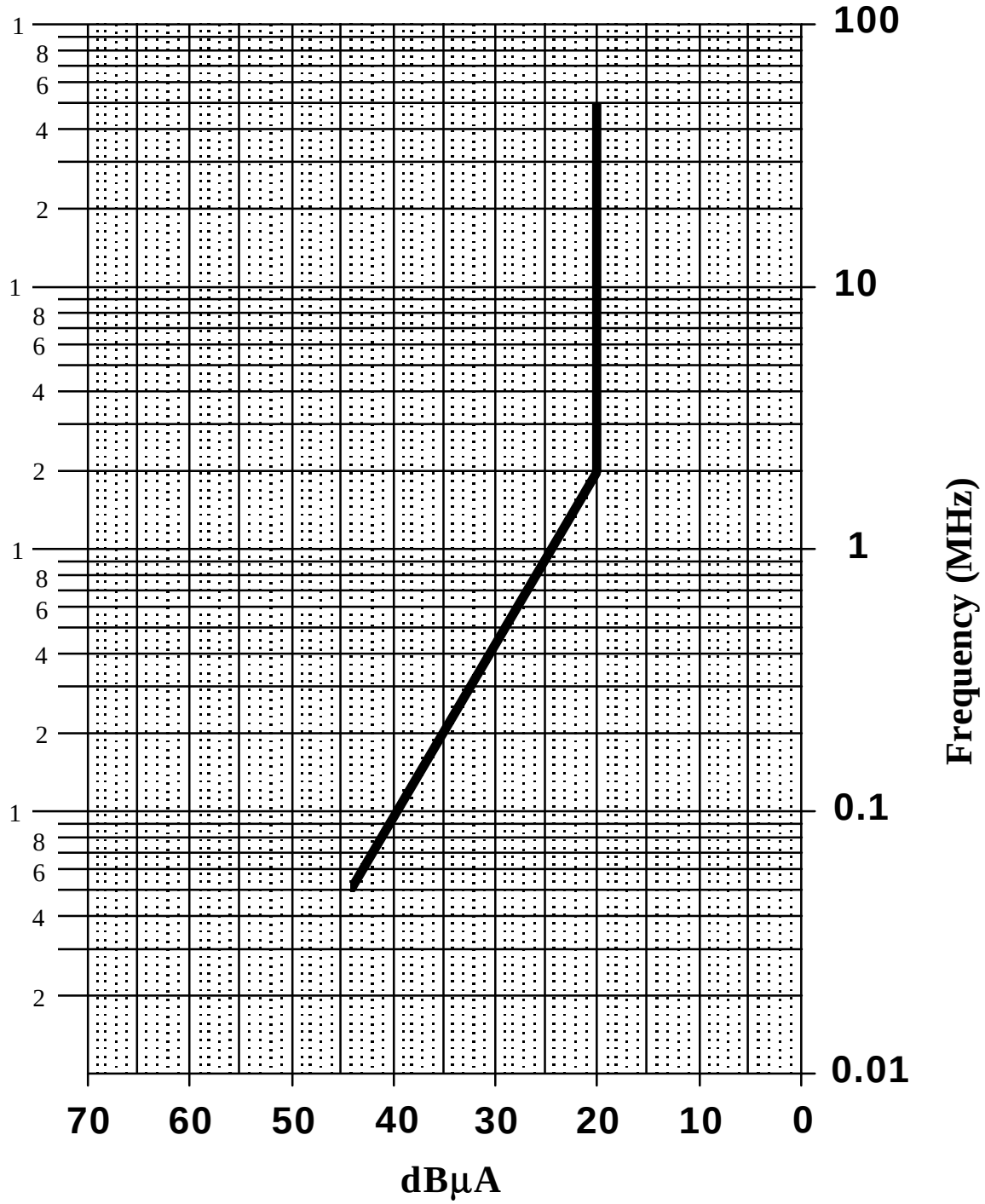


FIGURE 5. Power lead electromagnetic emission limits.

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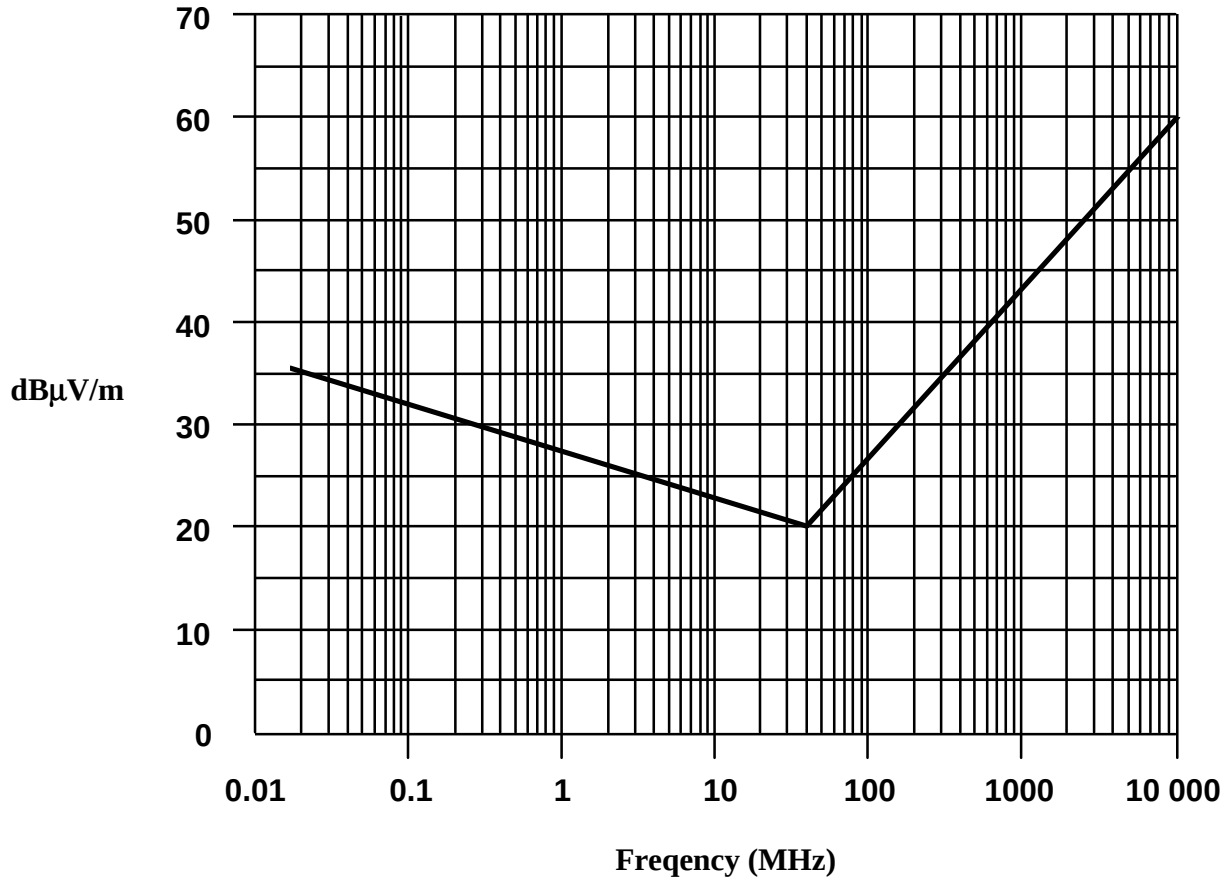


FIGURE 6. Field emission limits.

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	MIL-PRF-62315C	970725

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HEATER ASSEMBLY, COMBUSTION VEHICULAR COMPARTMENT, 60 000 BTU/HR

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

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c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	(810) 574-8745 786-8745 IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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